
Velib data Analysis with python

Final Report

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Abstract

The *Velib'* service implemented by the City of Paris is a privileged observatory of soft urban mobility in the French capital. By combining open real-time Velib' availability data with topographical and geographical information, this paper aims to analyze the dynamics driving station traffic according to the time of day. It lays the foundations for the implementation of a predictive model, enabling service improvements to make shared bicycle usage more fluid and comfortable for users.

The data used was acquired over the period of November and December 2025.

In this study, we utilize tools for *statistics*, *modeling*, and *cartographic visualization*. This article constitutes the final assignment for the Advanced Data Analysis with Python course taught during the first semester of the 2025/2026 academic year at Sciences Po Paris. The GitHub repository for the project is available via this [link](#).

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1 Introduction

1.1 The bicycle: a mode of transport for Parisians

In Paris, the bicycle is a tool used for daily commuting. While its usage has risen sharply over the last 50 years, Parisians have always had a strong relationship with this type of mobility. Indeed, as early as 1910, the Massard report [8] recorded an increase in bicycle usage between 1898 and 1908 (a 2.8-fold increase). Starting from 1905, the velodrome concluded the final stage of the Tour de France, continuing until the first finish on the Champs-Élysées in 1975.

While this phenomenon is not new, it has been marked by a sharp increase since 1970 (Figure 1). In its latest study on the development of cycling in the Greater Paris metropolis [4], the Parisian Urban Planning Agency (APUR) explains this increase by the concurrence of two public policies: the development of cycling infrastructure and the implementation of a self-service bicycle system. Indeed, the number of bicycles has continued to grow in the Greater Paris metropolis, and even more strongly within the municipality of Paris. The redevelopment of the riverbanks in 2010, presented by the City of Paris as a process of "reconquest of the Seine banks by pedestrians and cyclists" [9], was of unprecedented scale. This allowed for the densification of the cycle lane network present in the capital.

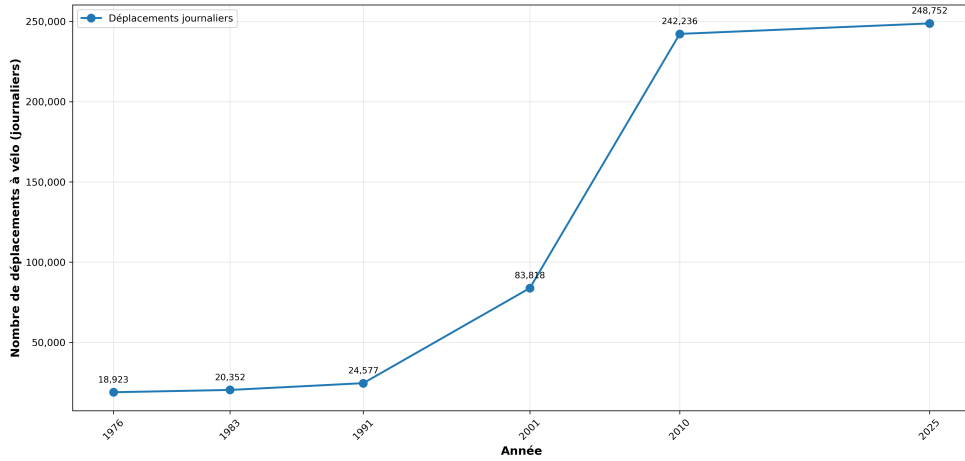


Figure 1: Evolution of daily bicycle usage in Paris

1.2 Velib': a catalyst for cycling practice

The second element is the creation of a public bike-sharing service: Velib'. APUR refers to Velib' as "a catalyst for cycling practice since 2007". Indeed, this service has greatly contributed to the boom in cycling, first in Paris and then in Greater Paris [5]. In 2024, this service accounts for 27% of bicycles riding on cycle paths. There are no fewer than 19,700 bicycles in the various Parisian stations, 40% of which benefit from electric assistance (e-bikes) [1].

Parisians or passing visitors can benefit from this service by subscribing to one of the three proposed plans. The total number of subscriptions has grown from 146,460 in 2018 to 461,770 active subscriptions in October 2024. This represents a threefold increase in subscriptions.

Thus, Velib' constitutes a structural change in the way people move around Paris. However, its practice, implementation, and operation are not homogeneous across the entire territory, and it is necessary to study its dynamics and quantify them.

1.3 Velib' stations: an object of analysis to make the service fluid and more comfortable for users

The analysis of Velib' data has already been carried out by some researchers to describe the status of stations in real-time using API querying [2]. This research led to the creation of a web application *Velib' Now!* [6], an application that locates the nearest available Velib' bikes or parking docks to your position. Others have focused on interactive geospatial visualization using Folium maps to visualize commuter bike flows [3]. This quantitative data approach should be related to qualitative surveys conducted by public institutions. Indeed, the Parisian Urban Planning Agency [4] published a comprehensive study on bicycle usage in Paris in October 2025. Part of their survey focuses on describing the typical Velib' user. It also informs us about the structure of the Velib' network and how it integrates into cycling infrastructure. Furthermore, the web media *Transition Vélo*, specialized in soft mobility issues [11], informs us about the context in which this Velib' revolution is taking place.

While these elements are fundamental to understanding the context of this study, we must also place it within broader academic research on constructing learning and prediction models. Indeed, while describing the situation of Velib' is necessary for understanding cycling, predicting its evolution is fundamental to maintaining a level of service. For this, approaches exist in data science such as time series construction. There are also more complex models in scientific literature that allow taking not only time as a variable but also spatial dependencies (Spatial-Temporal Graph Convolutional Networks (ST-GCN)). This is what Guanyao Li studies, for example, in his article: "a Data-Driven Spatial-Temporal Graph Neural Network for Docked Bike Prediction [7]".

The work we present in this article does not content itself with a simple description of the mechanisms at work in the filling of Velib' stations but lays the foundations for the implementation of a predictive model, to allow agents to anticipate station dynamics with the objective of making shared bicycle usage more fluid and comfortable for users.

Thus, we will focus on the dynamics of Parisian stations, centering on the following research question: **Can we anticipate the Velib' flows that cause stations to fill up or empty out?**

2 Methodology

This project aims to analyze the spatial and temporal distribution of Velib' stations in Paris, with a particular focus on the impact of altitude on station occupation and the distribution of electric versus mechanical bicycles. The central hypothesis is that altitude could influence usage behaviors and therefore the availability of different types of bikes.

Specific objectives of this study include:

- Mapping the distribution of Velib' stations across the Parisian territory
- Analyzing the influence of altitude on station occupation
- Studying the distribution of electric bicycles according to altitude and station clusters
- Developing predictive models (ARIMA/SARIMA) to anticipate availability
- Identifying station typologies via spatial clustering

2.1 Data Acquisition

2.1.1 Data Availability

Our work focuses on the situation of Velib' in the city of Paris. It is therefore appropriate to first collect the data in order to process it.

Although the data is available as open data, the City of Paris Data Lab does not have historical data records. The datasets are available and queryable via an API that authorizes the search and download of records with several parameters but only in real-time.

The collected data corresponds to 34 CSV files covering the period from November 13 to December 16, 2025, with hourly readings for each station. This longitudinal approach allows capturing temporal variations (daily cycles, differences between weekdays and weekends) essential for understanding the dynamics of the Velib' system. Finally, this allows for the constitution of a consolidated dataset of over 500,000 observations with key variables: Station Code, Timer, Bikes Available, Ebikes, Mechanical Bikes, Capacity, Geographical Coordinates.

2.1.2 Data Import

We implemented a protocol to retrieve Velib' data from the Paris Data API and save it to a CSV file on an hourly recurring basis [10].

This automation was set up using GitHub Actions, which allows launching a python file `velib_fetch.py` regularly (every hour) to call the Velib' API. These files are "committed" to GitHub in daily files.

The `'velib_fetch.py'` script allows making an API call. This script performs an HTTP GET request to check the API's availability. The script can thus access the `'records'` key of the JSON and extract real-time data for each station.

For each station, we can extract key information such as the identifier, the number of electric and mechanical bikes available, available spots, total capacity, station location, and the time of the request. The script thus allows structuring these acquisitions into a dataframe which it transforms into CSVs that are saved to the GitHub project directory following a format `'data/velib_data_YYYY-MM-DD.csv'`. We can then read the CSV files present in the GitHub directory from our script to extract the geographical coordinates of each station from the string format into distinct numeric latitude/longitude columns. We then perform spatial filtering by eliminating observations without valid coordinates and verifying geographical consistency (whether stations are effectively located in Paris). Finally, we create geometric objects by converting our data into a GeoDataFrame with Point objects to facilitate subsequent spatial analyses.

2.2 Data Processing

2.2.1 Data Display

During this study, we produced several visualizations. Initially, the instantaneous distribution of bicycles was mapped statically, by superimposing stations (sized according to their stock) on an OpenStreetMap background, allowing for the identification of immediate tension zones.

To go beyond this frozen view and capture daily cycles, we proceeded with a spatial aggregation of data by administrative districts (arrondissements). This processing served as the basis for producing an interactive animation exported in HTML format, generated via the `FuncAnimation` function of the Matplotlib library. This dynamic support allows visualizing, hour by hour, the evolution of occupation rates (normalized on a scale of 0 to 100%), thus visually revealing urban commuter movements and the beating heart of Paris. In this sense, we clearly observe a morning depletion of peripheral zones and saturation during the day in the city center, which we would not have been able to see on a classic static map.

2.2.2 Libraries and Tools Used

Data processing was carried out entirely in Python. After downloading the data, structuring and cleaning mobilized the Pandas library. We used the notebook format (.ipynb) for code development, and the python format (.py) for its execution.

We were then able to rely on GeoPandas and SciPy to perform spatialization and cross-referencing of stations with the Digital Elevation Model (via bilinear interpolation).

The analysis and production of dynamic visualizations were carried out with Matplotlib.

Regarding modeling, we used the statistical tool Statsmodels for the development of temporal predictions (SARIMA).

Finally, station segmentation by clustering (K-Means) and random forests were made possible thanks to the Scikit-learn tool.

3 Results and Discussion

3.1 Geographical and Altimetric Contextualization

3.1.1 Integration of the Digital Elevation Model (DEM)

We chose to use the Paris DEM at 25m resolution to finely characterize the altitude of each station. From there, we apply a bilinear interpolation via `RegularGridInterpolator` to precisely estimate the altitude of each station from its Lambert-93 coordinates. This approach allows going beyond the limits of a simple "high zone/low zone" classification by precisely quantifying elevation changes, which are potentially determining factors in the choice between electric and mechanical bicycles.

3.1.2 Visualization on Parisian Relief

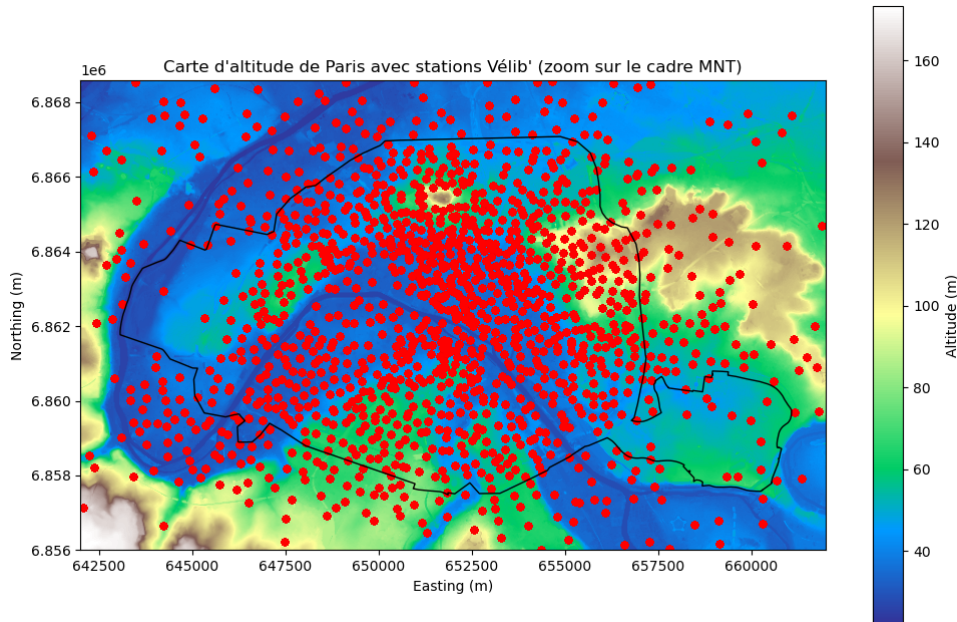


Figure 2: Altitude map with Velib stations

This first visualization (Figure 2) reveals the heterogeneous distribution of stations across the Parisian territory. A notable concentration is observed in central zones (moderate altitude 30-60m) and a more dispersed presence on the reliefs of Montmartre, Belleville, and Ménilmontant (up to 120m altitude). We then observe that the distribution of stations does not perfectly follow

the topography, suggesting that other factors (urban density, population flows) predominate in the network's implementation.

3.1.3 Statistical Distribution of Altitudes

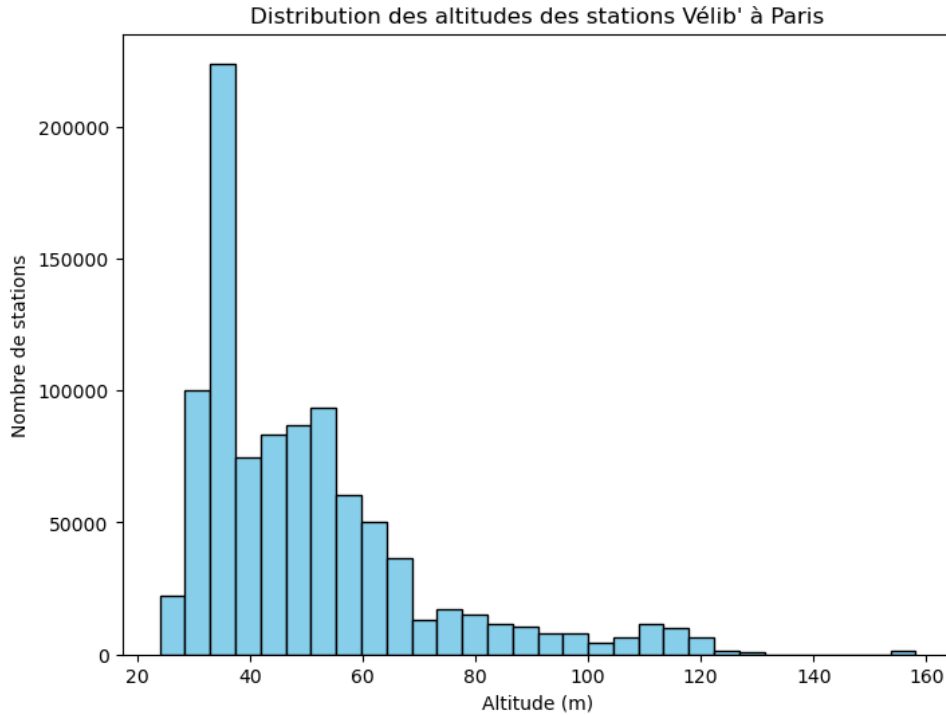


Figure 3: Histogram of station altitudes

Statistical analysis reveals different representative values:

- Mean altitude: 50.17 m
- Median altitude: 44.85 m
- Standard deviation: 20.48 m

This distribution (Figure 3), slightly skewed towards higher values, confirms that the majority of stations are located in the central part of Paris (30-70m), with a distribution tail towards the heights (Montmartre, Buttes-Chaumont).

3.2 Temporal Predictive Modeling

3.2.1 Attempt at Classic ARIMA Modeling

We randomly choose a station, in this case, Caumartin-Provence (Code 9104), which is a representative station of central Paris.

To prepare the data, we perform:

- Hourly resampling to create a regular time series
- Data differentiation to ensure stationarity

- A train/test split (90%/10%)

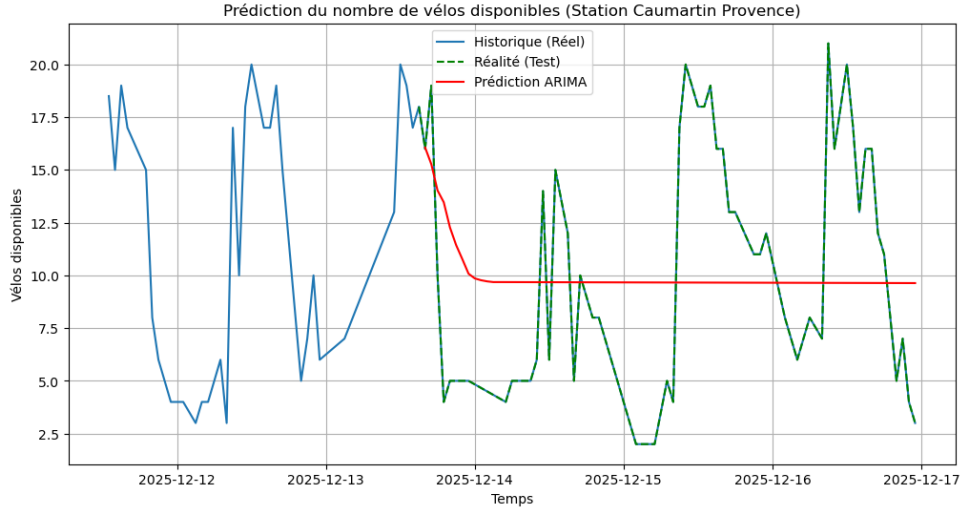


Figure 4: ARIMA model predictions for Caumartin Provence station

As shown in Figure 4, the ARIMA model (1,0,12) shows disappointing performance, unable to capture marked daily cyclical patterns.

We see that ARIMA predictions remain too smoothed and do not reproduce the significant variations linked to usage cycles (morning peaks, nocturnal troughs). This limitation justifies the switch to a seasonal model, such as SARIMA.

3.2.2 Implementation of the SARIMA Model

We retain the SARIMA(1,0,1)(1,1,0,24) model with a 24-hour seasonal cycle.

This configuration is justified for several reasons:

- $s=24$: explicit recognition of the daily cycle
- $D=1$: seasonal differentiation to eliminate daily trends
- $(P,Q)=(1,0)$: correlation with the same time the previous day

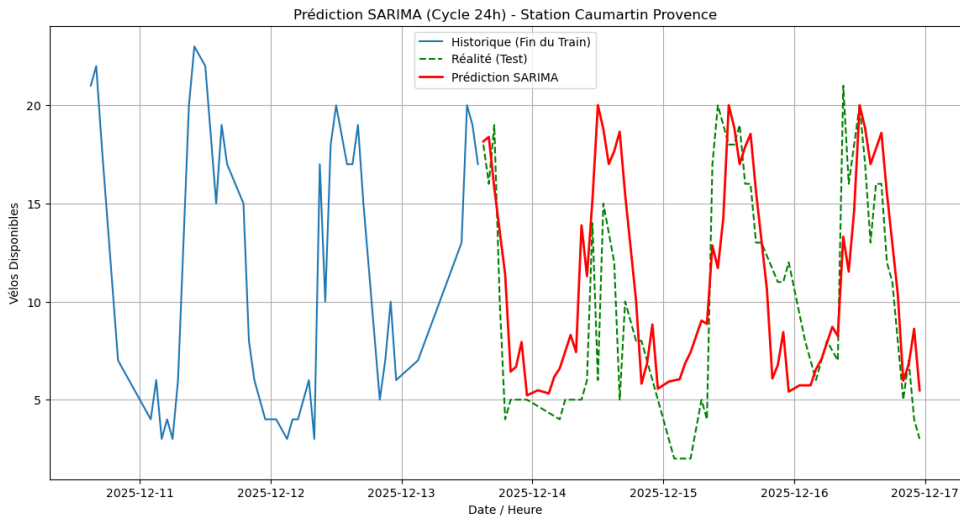


Figure 5: SARIMA Predictions vs Reality

The SARIMA model captures daily oscillations much better. We observe (Figure 5) a faithful reproduction of morning patterns (depletion around 8am-9am) and nocturnal replenishments. This improvement validates the importance of the seasonal component in modeling Velib flows.

Here, the recognition of urban usage cycles was crucial for prediction and explains our transition from ARIMA to SARIMA.

3.3 Dynamic Cartographic Analysis

3.3.1 Static Mapping by Moment

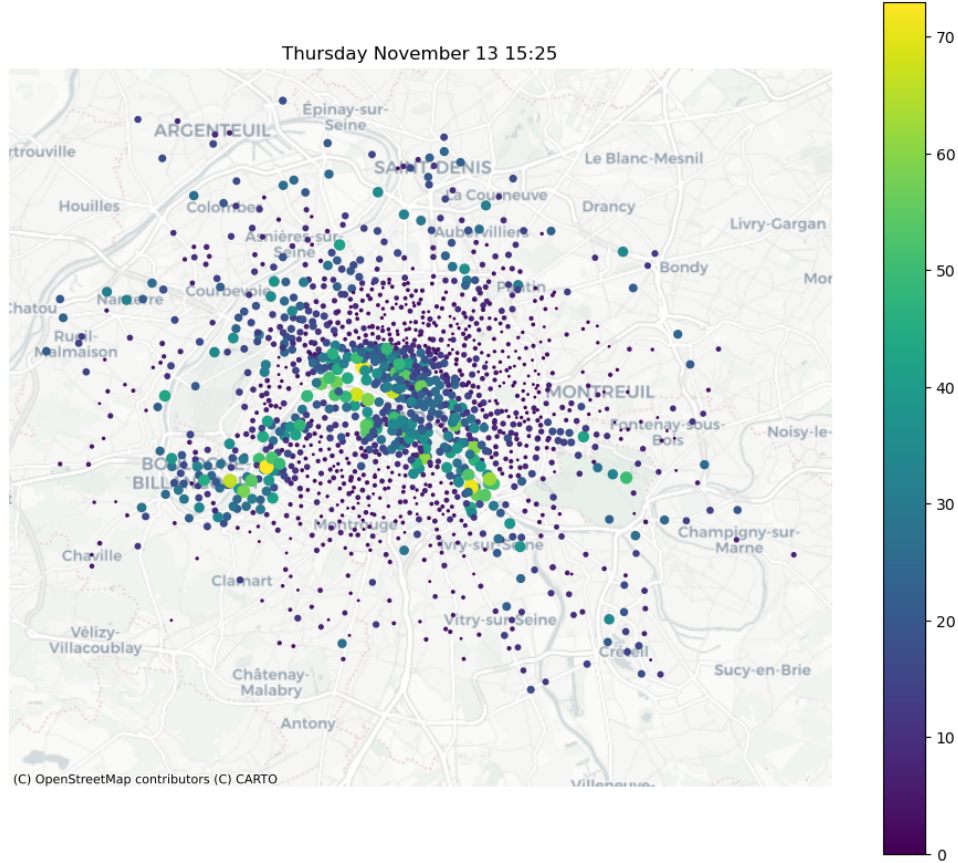


Figure 6: Instantaneous distribution of available bicycles

In order to observe the instantaneous distribution of available bicycles throughout Paris, we superimpose the stations (size proportional to the number of bikes) on an OpenStreetMap background.

This map (Figure 6) reveals spatial disparities in occupation at a given moment. Central zones generally show lower availability (high rotation), while peripheral zones maintain larger stocks.

3.3.2 Analysis by Arrondissements

We perform a spatial aggregation of data by Parisian neighborhoods via a geospatial join, allowing us to calculate average occupation rates by administrative zone. This will allow us to create an animation showing the hour-by-hour evolution of availability in the different districts of Paris (Figure 7).

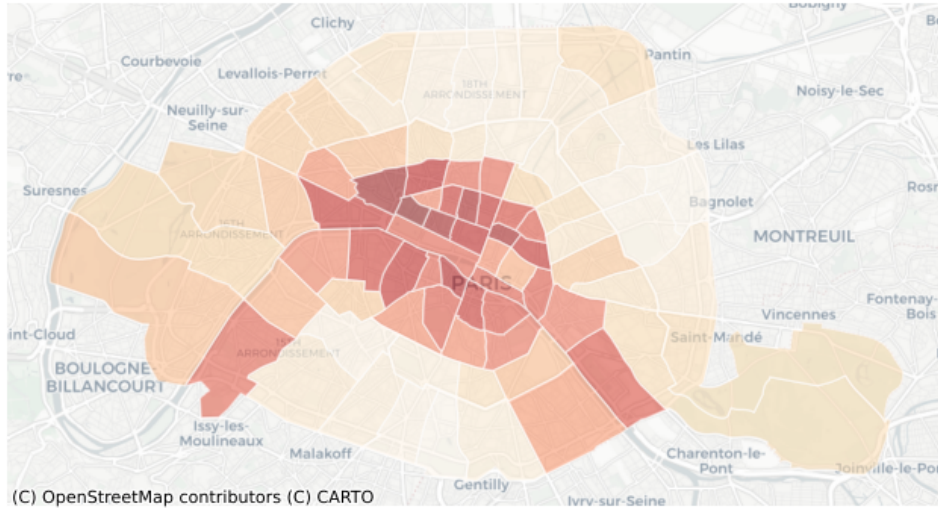


Figure 7: Occupation rate by neighborhood at a given moment

The choropleth map reveals heterogeneous occupation patterns: central districts (1st, 2nd, 3rd, 4th) display strong occupation rates while peripheral residential districts maintain lower stocks. This geography of occupation reflects home-work flows and differences in usage between activity zones and residential zones.

3.3.3 Temporal Animation

We then decide to create a FuncAnimation showing the hourly evolution of occupation rates, with a normalized colorbar (0-100%).

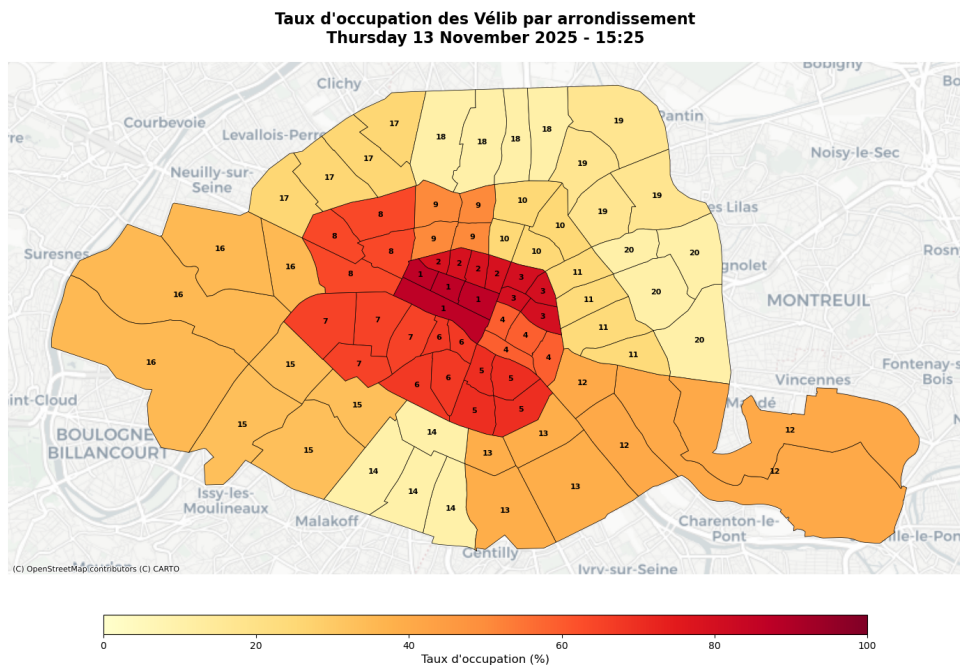


Figure 8: Image extracted from the animation of occupation rate by district

The animation is available via this [link](#). Please wait a few seconds for the frames to load! Figure 8 reveals clear cycles:

- **Mornings (7am-9am):** Progressive depletion of residential districts (15th, 16th, 19th, 20th)
- **Daytime (10am-5pm):** Concentration in the center (1st-4th districts)
- **Evenings (6pm-9pm):** Movement towards the Marais (3rd and 4th district), where people go out a lot
- **Night (10pm-6am):** Replenishment of peripheral stocks

These patterns confirm the hypothesis of home-work pendulum flows as the main determinant of the spatial distribution of bicycles. We clearly see 2 distinct zones of influence and attraction: the center and the peripheral residential zones.

3.4 Spatial Clustering and Station Typology

3.4.1 Construction of Characterization Variables

We wish to perform spatial clustering to validate the hypothesis of 2 zones, of two types of stations: central and residential.

We perform a statistical approach by aggregating behavioral metrics by station:

- **occupation_mean:** average occupation rate
- **occupation_std:** variability of occupation
- **pct_empty:** percentage of time empty
- **pct_full:** percentage of time saturated
- **mean_capacity:** average capacity

These indicators synthesize the temporal behavior of each station, allowing for the identification of distinct usage profiles beyond simple geographical location.

3.4.2 Determination of Optimal Number of Clusters

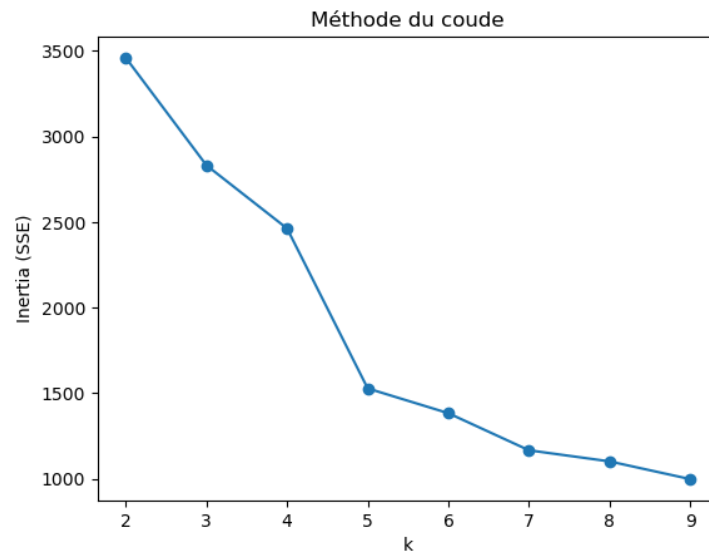


Figure 9: Elbow Method

Inertia analysis suggests optimal segmentation into 2 clusters (Figure 9), with a marked elbow at $k=2$. This bipartition likely reflects the fundamental opposition between central zones (high rotation) and peripheral zones (residential usage).

3.4.3 Cluster Profile Analysis

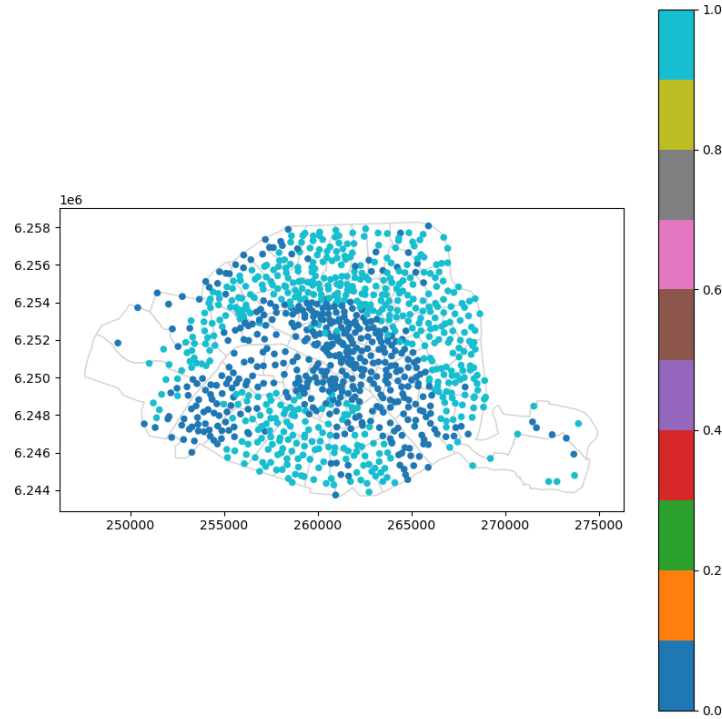


Figure 10: Cartography of clusters

The spatial distribution of clusters confirms the hypothesis of a radioconcentric organization of the Velib system, with a central heart of high rotation and peripheral zones of more residential usage (Figure 10).

3.4.4 Time Series Analysis by Cluster

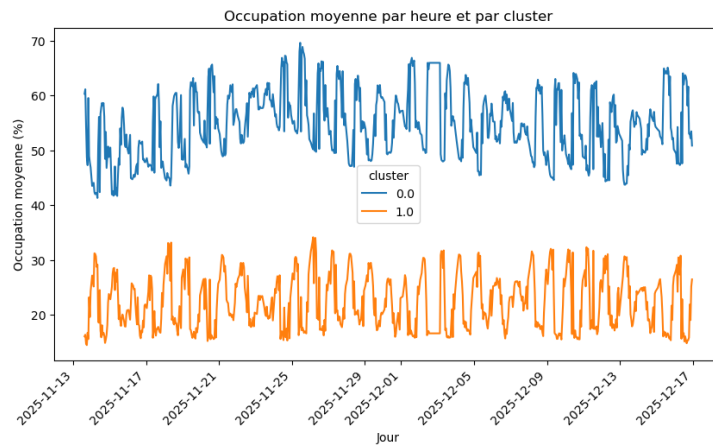


Figure 11: Average temporal evolution by cluster

We see that temporal profiles (Figure 11) confirm the functional bipartition:

- **Cluster 0:** Stations in the center of Paris are constantly 50-60% full. They fill up during the day and empty in the evening.
- **Cluster 1:** Residential stations are constantly 20-30% full. They empty in the morning and fill up in the evening.

3.5 Analysis of Altimetry Influence

3.5.1 Modeling Altitude Effect on Global Occupation

We wish to analyze the impact of altitude on the distribution of Velibs in Paris. For this, we first use a mixed model with random effects by station to control for unobserved heterogeneity:

```
occupation ~ altitude_m + cluster + (1|Code Station)
```

Statistical results show us that:

- The altitude coefficient is weakly positive but not significant
- Altitude does not significantly explain variations in occupation
- The cluster effect remains the main determinant

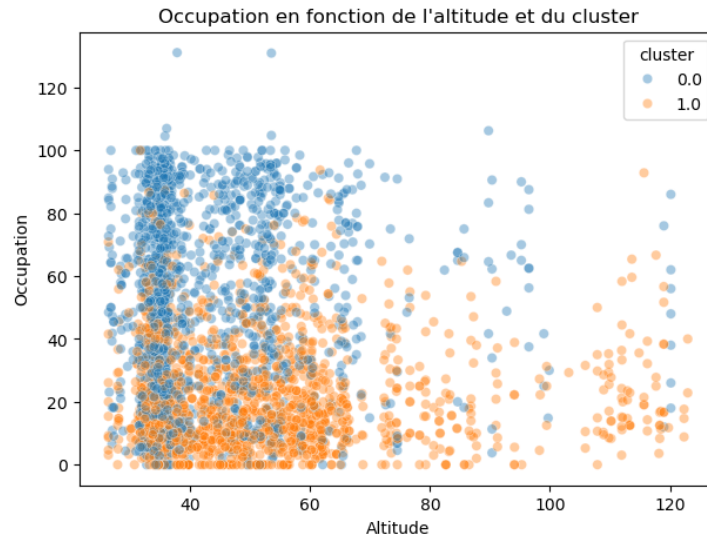


Figure 12: Occupation vs Altitude Scatter Plot

In Figure 12, the significant dispersion of points and the absence of a clear trend confirm that altitude is not a determining factor in general station occupation. Clusters (center/periphery) remain the dominant explanatory variable.

3.5.2 Specific Analysis of Electric Bicycles

We test the hypothesis that altitude could specifically influence the demand for electric bicycles (assistance for climbs).

The variable of interest we use is: $\text{ratio_ebikes} = \text{Ebikes} / \text{Capacity}$

Results from the mixed model show that:

- altitude-ebike ratio correlation: very weak (< 0.05)
- the apparent importance of altitude in the Random Forest (94%) is certainly a statistical artifact (cardinality bias)

Contrary to intuition, altitude does not significantly determine the distribution of electric bicycles. This absence of effect can be explained by several factors, such as the rapid rotation of bikes preventing accumulation in certain areas, the algorithmic management of the Velib system which actively redistributes bikes, or the limited altimetric range of Paris (90m difference) which is insufficient to create discriminatory effects.

3.5.3 Fine Temporal Analysis by Altitude Quartiles

We still attempt to analyze the distribution of electric Velibs by altitude quartile within each cluster separately (Figure 13 and 14).

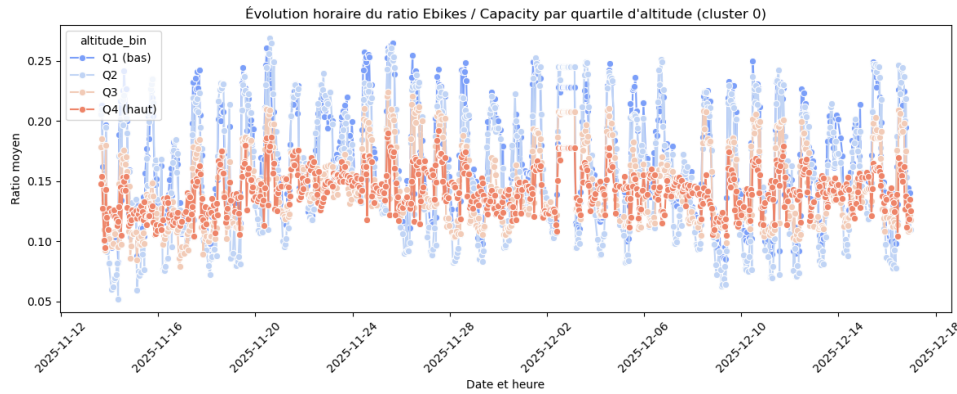


Figure 13: Hourly evolution of electric/mechanical ratio by altitude (Cluster 0)

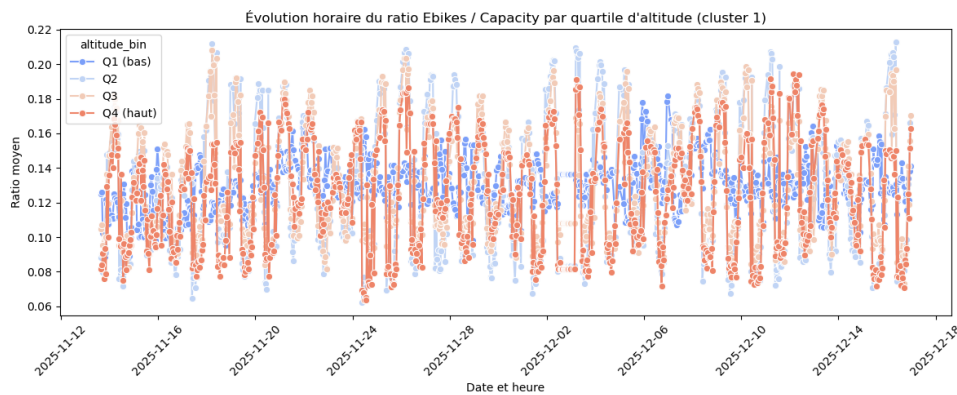


Figure 14: Hourly evolution of electric/mechanical ratio by altitude (Cluster 1)

We segment stations into altitude quartiles (Q1-Low to Q4-High) and track the hourly evolution of the electric ratio.

For cluster 0 (center), we see (Figure 13) superimposed curves between altitude quartiles, an electric share stable around 15%, and no discriminatory effect of altitude (probably because center stations have very low altitude variation).

For cluster 1 (residential), we see (Figure 14) greater variability but without a coherent altimetric pattern. Intermediate quartiles (Q2, Q3) sometimes show the highest ratios, and there is an absence of systematic domination by high altitudes.

These results definitively invalidate the hypothesis of an altitude effect. The electric/mechanical distribution seems rather determined by local demand factors and system management.

3.6 Aggregated Analysis of Electric Distribution

3.6.1 Spatial Aggregation Methodology

Station-by-station ratios can be biased by empty stations or those with very low capacity, which is why we attempt to adapt our study by performing aggregation by altitude groups before calculating the ratio. This approach avoids over-weighting small stations and provides a "macro" view of electric availability by altitude zone.

3.6.2 Final Results by Cluster and Altitude

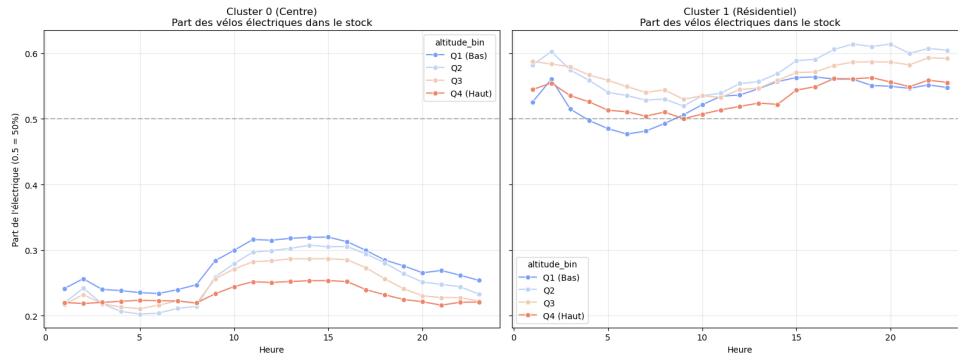


Figure 15: Share of electric bicycles in total stock

Figure 15 presents the share of electric bicycles in the total stock for the two defined clusters.

For cluster 0 (center), we see that the share of electrics is generally lower, around 20-30%. We reconfirm the absence of an altitude effect between quartiles.

For cluster 1 (residential), the electric share is generally higher (50-60%). There is variability between altitude quartiles but without a systematic pattern. Intermediate zones (Q2, Q3) seem slightly more "electrified".

We can thus conclude that the major opposition lies between clusters (center vs periphery) and not between altitude levels. Residential zones have a larger share of electric bicycles, likely linked to longer trips.

4 Conclusion

The first major finding of this study lies in the validation of a spatial organization dictated by the functionality of the Parisian territory. The data confirms that the network is structured by a profound duality between central zones of high rotation and residential peripheral zones, where usage proves to be significantly more stable. Conversely, the analysis surprisingly invalidates the influence of altitude. Contrary to expectations, Parisian relief exerts no significant impact on station occupation nor on the distribution between electric and mechanical bicycles, suggesting that the physical effort linked to elevation gain is neutralized by other factors.

These conclusions open up concrete operational perspectives. To optimize system performance, it appears more strategic to prioritize rebalancing between activity hubs and residential areas rather than focusing on relief zones. Similarly, the maintenance and distribution of the electric fleet should adjust to functional needs, such as long-distance trips, rather than solely to topography. In terms of modeling, these results invite prioritizing SARIMA-type models integrating daily seasonality and favoring socio-economic variables as principal predictors of demand.

Finally, while this research provides solid answers, it also paves the way for new questions. The observation period, limited to just over a month, would merit extension to integrate seasonal and meteorological variations. The lack of data on actual trips constitutes another limitation that a future analysis of origin-destination flows could fill. Ultimately, integrating variables linked to public policies and major urban events would allow for even further refinement in understanding this system, while opening the way for comparisons with other global metropolises.

By mobilizing several methodological tools, combining geostatistics, machine learning clustering, temporal modeling, and mixed-effects models, we were able to reveal the real complexity of the Velib' system. This approach demonstrated that functional factors and algorithmic mechanisms largely dominate geophysical determinants in the organization of the service.

The analysis thus underscores the crucial importance of choosing observation scales in interpreting urban phenomena. It is by articulating the scale of the individual station with vaster spatial aggregates and by confronting the immediacy of data with their temporal evolution that the deep logics of shared mobility emerge.

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